

UNIT - 4 STATISTICAL INFERENCE :-

→ The methods of statistical inference are divided into two major cases

a, Estimation

b, Test of Hypothesis → (guess or assumption)

Estimate:- To find an unknown population parameter, a judgement or a statement is made which is called as "Estimate".

Statistical Estimation:- It is a part of statistical inference where a population parameter is estimated from the corresponding sample.

Point Estimation:- Point estimation of a parameter is the estimation where a population parameter is estimated by using a single numerical value from the corresponding sample data. If we use the value of the sample statistic to estimate the population parameter, then that value is called as "Point Estimate".

Estimator:- The statistic whose value is used as a point estimate of the parameter is called as an "Estimator".

Eg:- If we use sample mean ' $\bar{x}$ ' to estimate population mean ' μ ' and sample variance to estimate population variance and so on, then this sample mean, sample variance and so on are called as Estimators.

Properties of a Good Estimator:-

1, Unbiasedness:-

An estimator ' $\hat{\theta}$ ' is said to be unbiased estimator of population parameter if $E(\hat{\theta}) = \theta$

if $E(\hat{\theta}) > \theta$, then $\hat{\theta}$ is said to be positively biased and if $E(\hat{\theta}) < \theta$, then $\hat{\theta}$ is said to be negatively biased and the amount of biasedness is given by,

$$b(\hat{\theta}) = E(\hat{\theta}) - \theta$$

2, Consistency:- If an estimator ' $\hat{\theta}$ ' approaches the parameter ' θ ' closer and closer as the size increases, then we say that ' $\hat{\theta}$ ' is consistent estimator.

3, Efficiency:- If the estimators are ' $\hat{\theta}_1$ ' and ' $\hat{\theta}_2$ ' for a parameter ' θ ', then ' $\hat{\theta}_1$ ' is said to be more efficient than ' $\hat{\theta}_2$ ' for all sample sizes, if $\text{var}(\hat{\theta}_1) < \text{var}(\hat{\theta}_2)$

4, Sufficiency:- An estimator is said to be sufficient for a parameter ' θ ', if it contains all the information in a sample regarding the parameter.

Maximum Error of Estimates:- Since the estimate of sample mean $\bar{x}$ is very rarely equals to the population mean μ , the difference b/w the estimator and the quantity to be estimated is given by the error

$$|\bar{x} - \mu|$$

→ For large samples, $\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ is a random variable having standard normal distribution, then the inequality

$$-Z_{\alpha/2} \leq \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2} \Rightarrow \frac{|\bar{x} - \mu|}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}$$

is satisfied with $(1-\alpha)$ 100% confident

→ Where $Z_{\alpha/2}$ represents the area under the normal curve on its right side equal to $\alpha/2$.

→ Let 'E' represents maximum error of $|\bar{x} - \mu|$, then

$$E = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

→ maximum error of estimates with probability $(1-\alpha)$

Note:- The most commonly used values for $(1-\alpha)$ are 0.95 and 0.99 and their corresponding values of $Z_{\alpha/2}$ are $Z_{0.025} = 1.96$ and $Z_{0.005} = 2.58$

Means:-

Max error of estimate:-

$$E = \begin{cases} Z_{\alpha/2} \frac{s}{\sqrt{n}} & \text{when } n \geq 30 \text{ (large sample)} \\ t_{\alpha/2} \frac{s}{\sqrt{n}} & \text{when } n < 30 \text{ (small sample)} \end{cases}$$

2, Sample size(n):-

$$E = Z_{\alpha/2} \frac{s}{\sqrt{n}} \Rightarrow n = \left(Z_{\alpha/2} \frac{s}{E} \right)^2$$

3, Confidence interval:-

$$\bar{X} - E < \mu < \bar{X} + E$$

4, Degree of Confidence:- Same formula we can use.

Note:- Mostly used (1- α) 100% values are 99% and 95%. $Z_{0.025} = 1.96$ $Z_{0.05} = 2.576$ or 2.58

Q, A random sample of size 100 has std of 5. What can you say about max error with 95% confidence.

Ans:- $n = 100$, $s = 5$ 95% confidence, $\alpha = 5\% \Rightarrow \frac{\alpha}{2} = 0.025$.

$$Z_{\alpha/2} = Z_{0.025} = 1.96$$

$$E = Z_{\alpha/2} \frac{s}{\sqrt{n}} = 1.96 \times \frac{5}{10} = 0.98$$

Q, what is maximum error one can expect to make with probability 0.9 when using the mean of random sample of size $n = 64$ to estimate mean of population with $\sigma^2 = 256$.

$n = 64$ (large) confidence = 90% $\alpha = 10\% \Rightarrow \frac{\alpha}{2} = 5\% = 0.05$.

$$s = \sqrt{\frac{256}{4}} = 16$$

$$E = Z_{\alpha/2} \frac{s}{\sqrt{n}} = 1.645 \times \frac{16}{8} = 3.29$$

Q, Assuming that $\sigma = 20$, how large a random sample to be taken to assert with probability 0.95 that sample mean will not differ from true mean by more than 3.0 points

$\sigma = 20$, Conf = 95% $\alpha = 0.025$

$$Z_{\alpha/2} \frac{20}{\sqrt{n}} \leq 3$$

$$\sqrt{n} > \frac{1.96}{3} \times 20$$

$$n > 170.7 \approx 171$$

Q, It is desired to estimate the mean number of hours of continuous use until a certain computer will first require repairs. If it can be assumed that $\sigma = 48$ hours, how large a sample be needed so that one will be able to assert with 90% confidence that sample mean is off by almost 10 hours.

Ans: $\sigma = 48$

$$\alpha = 0.025$$

$$Z_{\alpha/2} = 1.96$$

$$1.96 \times \frac{48}{\sqrt{n}} \leq 10$$

$$\sqrt{n} > \left(\frac{48}{10} \times 1.96 \right)^2 > 62.3$$

$$n = 63$$



Q, A random sample of size 81 was taken $\sigma^2 = 20.25$, mean = 32. Construct 98% confidence interval

Ans: $n=81$ $\sigma^2=20.25 \Rightarrow S=4.5$, $\bar{X}=32$

$\alpha=0.01$ $Z_{0.01} = 2.326$

$E = 2.326 \times \frac{4.5}{\sqrt{81}} = 1.163$

confidence interval is given by

$\bar{X} - E < \mu < \bar{X} + E$

$30.837 < \mu < 33.163$

Q, A random sample of size 100 is taken from a population with $\sigma=5.1$ and sample mean $\bar{X}$ is 21.6. Construct 95% confidence interval for population mean μ

Ans: $n=100$ $Z_{\alpha/2} = 1.96$ $\sigma = 5.1$

$E = 1.96 \times \frac{5.1}{10} = 0.9996$

$\bar{X} - E < \mu < \bar{X} + E$

$20.6004 < \mu < 22.5996$

Q, Find 95% confidence limits for mean of normally distributed population from which sample was taken: {15, 17, 10, 18, 16, 9, 7, 11, 13, 14}

Ans: $\bar{X}=13$ $Sf = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}} = \sqrt{\frac{4+16+9+25+9+16+36+4+0+1}{9}}$, $n=10$ (small)

$S = \frac{10.95445}{10} = 3.651$ $\alpha = 0.025$

$E = 1.96 \times \frac{3.651}{10} = 2.385$ $t_{0.025, 9} = 2.262$

$2.262 \times \frac{3.651}{\sqrt{10}} = 2.6115$

$\bar{X} - E < \mu < \bar{X} + E$

$10.389 < \mu < 15.611$

Q, The mean of the random sample is an unbiased estimate of mean of population $\{3, 6, 9, 15, 27\}$ then

- i) List all possible samples of size 3 that can be taken without replacement.
- ii) Calculate mean of each of samples listed in (i) and assigning each sample a probability of $\frac{1}{10}$. Verify that mean of these $\bar{X} = 12$ which is equal to mean of population μ i.e. $E(\bar{X}) = \mu$ i.e. prove that $\bar{X}$ is an unbiased estimator of μ

$N=5$ $n=3$ $x_i's \rightarrow 3, 6, 9, 15, 27$ No. of samples of size 3 without replacement = ${}^5P_3 = 10$

population mean $\mu = 12$ $(3, 6, 9), (3, 6, 15), (3, 6, 27), (3, 9, 15), (3, 9, 27), (3, 15, 27), (6, 9, 15), (6, 9, 27), (9, 15, 27), (6, 15, 27)$

$x = 6, 8, 12, 9, 13, 15, 10, 14, 16, 17$

$\sum \bar{X} = 120$ $\sum \bar{X}_i = 12$

$E(\bar{X}) = 12 = \text{population mean}$

$\bar{X}$	6	8	9	10	12	13	14	15	16	17
$f(x)$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$

Q. A random sample of 10 boys had the following IQs: 70, 120, 101, 88, 83, 95, 98, 107 and 100. Find reasonable range in which most of mean IQ values of samples of 10 boys lie. → Can use 99% or 95% conf

$$\bar{X} = \frac{70+120+110+101+88+83+95+98+107+100}{10} = 97.2$$

$$\text{Sample std} = s = \sqrt{\frac{739.84 + 519.84 + 163.84 + 144 + 184.64 + 201.64 + 4.84 + 0.64 + 2.44}{9}}$$

$$s = 14.2735$$

$$\frac{s}{\sqrt{n}} = 0.0005 \quad t_{\alpha/2} = 3.250$$

$$\frac{3.250}{\sqrt{10}} \times 14.2735 = 14.691$$

$$\bar{X} - E < \mu < \bar{X} + E$$

$$82.531 < \mu < 111.869$$

Degree of Confidence problems for mean:-

- Q. The efficiency expert of a computer company tested 40 engineers to estimate the average time it takes to assemble a certain computer component, getting a mean of 12.73 minutes and std of 2.06.
- If $\bar{X} = 12.73$ is used as a point estimate of actual average time required to perform the task determine max error with 99% confidence
 - Construct 98% confidence interval for true avg time it takes to do the job
 - With what confidence can we assert that sample mean doesn't differ from true mean by more than 30 sec.

Ans:- $n = 40$ $\bar{X} = 12.73 \text{ min}$ $S = 2.06 \text{ min}$

(i) $\alpha/2 = 0.005$ $Z_{0.005} = 2.576$ $= 2.576 \times \frac{2.06}{\sqrt{40}} = 0.839$

(ii) $\alpha/2 = 0.01$ $Z_{0.01} = 2.326$ $= 2.326 \times \frac{2.06}{\sqrt{40}} = 0.757$

$\bar{X} \pm E < \mu < \bar{X} + E \Rightarrow 11.973 < \mu < 13.487$

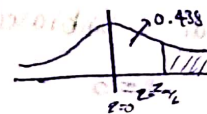
(iii) $E = 0.5$ $Z_{\alpha/2} \times \frac{2.06}{\sqrt{40}}$

$$Z_{\alpha/2} \leq \frac{0.5 \times \sqrt{40}}{2.06}$$

$Z_{\alpha/2} \leq 1.535$ $\alpha = 0.05$ $\alpha/2 = 0.025$
(see in t distribution table)

~~98% confidence~~ 1.534

$$\int_0^{1.534} f(z) dz = 0.438$$



degree of conf = $0.438 + 0.438 = 0.876 = 87.6\%$

Q. In a study of automobile insurance a random sample of 80 body repair cost had mean of ₹ 472.36 and std of ₹ 62.35. If $\bar{X}$ is used as a point estimate to true avg repair cost with what confidence we can assert that max error doesn't exceed ₹ 10.

Ans:- $n = 80$ (large) $\bar{X} = 472.36$ $S = 62.35$

$$E < 10$$

$$10 > Z_{\alpha/2} \times \frac{62.35}{\sqrt{80}}$$

$$Z_{\alpha/2} < \frac{10 \sqrt{80}}{62.35}$$

$$\int_0^{1.43} f(z) dz$$

$$\alpha/2 = 0.1236$$

$$Z_{\alpha/2} < 1.434$$

$$\alpha = 0.2472 \Rightarrow 1 - \alpha = 0.7528 \Rightarrow 75.28\% \text{ confidence}$$

Q, A sample of 11 rats from a central population had an avg blood viscosity of 3.92 with std of 0.61. Estimate 95% conf limits for mean blood viscosity of population

Ans: $n=11$ ($\Rightarrow$ small) $\bar{x} = 3.92$ $s = 0.61$
 $\alpha/2 = 0.025$ $t_{\alpha/2} = 2.228$

$$2.228 \times \frac{0.61}{\sqrt{11}} = 0.4097$$

$$3.92 - 0.4097 < \mu < 3.92 + 0.4097$$

$$3.5103 < \mu < 4.3297$$

Estimations of Proportions $\rightarrow$ standard error of proportion.

$$E = \begin{cases} Z_{\alpha/2} \sqrt{\frac{pq}{n}} & \text{for large sample} \end{cases}$$

$$\begin{cases} t_{\alpha/2} \sqrt{\frac{pq}{n}} & \text{for small sample} \end{cases}$$

$$\text{Sample size } n = \left(\frac{Z_{\alpha/2}}{E} \right)^2 \bar{p}\bar{q}$$

$$\bar{p} - E < p < \bar{p} + E$$

$$\bar{p} = \frac{x}{n}$$

$$\bar{q} = 1 - \bar{p}$$

Q, In a random sample of 160 workers exposed to certain amount of radiation, 24 experienced some ill effects. Construct 99% conf interval for corresponding ~~the~~ the.

~~n=160~~ $\alpha/2 = 0.005$ $Z_{\alpha/2} = 2.576$

Note: If $\bar{p}$ and $\bar{q}$ are not given then take $\bar{p} = \bar{q} = 1/2$
 Note: If degree of conf is not given then in proportions replace $Z_{\alpha/2}$ with 3

x = no. of people experiencing ill effects of radiation $\Rightarrow 24$

$x = 24$, $n = 160$ (large)

Sample proportion: $\bar{p} = \frac{x}{n} = \frac{24}{160} = 0.15$ $\bar{q} = 0.85$

$$E = 2.576 \sqrt{\frac{0.15(0.85)}{160}} = 0.0282 \times 2.576 = 0.0727$$

$$\bar{p} - E < p < \bar{p} + E$$

$$0.0773 < p < 0.2227$$

Q, What is size of smallest sample required to estimate an unknown proportion to within a max error of 0.06 with atleast 95% confidence.

$$Z_{0.025} \times \sqrt{\frac{\bar{p}\bar{q}}{n}} < 0.06$$

$$1.96 \times \sqrt{\frac{0.025}{n}} < 0.06$$

$$n = 267$$

$$n > \frac{0.025}{9.37 \times 10^{-4}} \Rightarrow n > 267$$



Q, In a sample of 100 packages shift by freight, 13 had some damage. Construct 95% confidence interval for true proportion of damaged package.

Ans $n = 100$ (large).
sample size

Let X represent no. of damaged package

$$X = 13$$

$$\alpha/2 = 0.025$$

Sample proportion $\bar{p} = \frac{13}{100} = 0.13$

$$Z_{0.025} \times \sqrt{\frac{0.13 \times 0.87}{100}} = 1.96 \times 0.065$$

$$\bar{p} - E < p < \bar{p} + E \Rightarrow 0.065 < p < 0.195$$

Test of Hypothesis Single mean large sample-

Q, An ambulance service claims that it can take on an avg less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 mins and var of 16 mins. Test significance at 0.05 level.

$n = 36$, $\bar{x} = 11$, $S^2 = 16 \Rightarrow S = 4$
Sample size (Large)

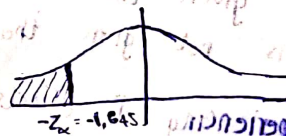
Null Hypothesis $H_0: \mu = 10$ (Against)

Alternate Hypothesis $H_1: \mu < 10$ (inferior) (L.T.D)

Level of significance $\Rightarrow \alpha = 0.05$. $Z_{0.05} = 1.645$

$$Z = \frac{\bar{x} - \mu}{S/\sqrt{n}} = 1.5$$

conclusion



Since z falls under acceptance region, accept null hypothesis.

$\therefore$ His claim is false.

Q, It is claimed that a random sample of 49 tyres has a mean life of 15,200 km. This sample was drawn from a population whose mean is 15,150 km and std of 1,200 km. Test significance at 0.05 level

Ans $n = 49$ (large) $\bar{x} = 15,200$ $\mu = 15,150$ $\sigma = 1,200$

$\alpha = 0.05$ $Z_{0.05} = 1.96$

N.H $H_0: \mu = 15,150$ (from pop)
A.H $H_1: \mu \neq 15,150$ (T.T.T)

$$Z = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{15,200 - 15,150}{1200/\sqrt{49}} = \frac{50 \times 7}{1200} = \frac{350}{1200} = \frac{7}{24}$$

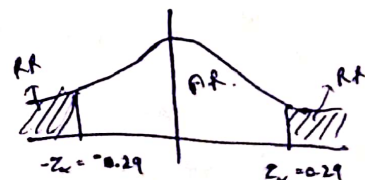
Null Hypothesis: $\mu = 15,150$ is taken from population

Alternate Hypothesis: $\mu \neq 15,150$ not taken (T.T.H).

$$\alpha = 0.05 \quad Z_{0.025} = 1.96$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{7 \times 50}{1200} = \frac{7}{24}$$

Accept Null hypothesis, His claim is True



Q, A ladies stenographer claims that she can take dictation at rate of 120 words/min. Can we reject her claim on basis of 100 trials in which she demonstrate a mean of 116 words and std of 15 words

Ans $n = 100$ (large)

$$\bar{x} = 116, s = 15$$

Null Hypothesis: $\mu = 120$

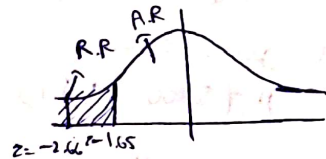
Alternate Hypothesis: $\mu \neq 120$ ($\mu > 120$ can also be used)
 $\propto$ (L.T.T) $\propto$ T.T.T

Let $\alpha = 0.05$

can also used $\alpha = 0.01$

$$Z_{0.05} = 1.645$$

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \Rightarrow Z = \frac{116 - 120}{15/\sqrt{100}} = -\frac{8}{3}$$



$z \in$ Rejection region

$\therefore$ Her claim is false.

Test of Hypothesis of single mean and small sample:-

Q, The heights of 10 nails of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is it reasonable to believe that average height is ≥ 64 inches? Test at 5% level of significance.

$n = 10$ (small) $\bar{x} = \frac{70+67+62+68+61+68+70+64+64+66}{10} = 66$

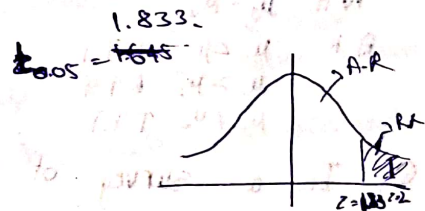
$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{16+1+16+4+9+25+4+16+4+0}{9}} = \sqrt{10} = 3.162$$

Null Hypothesis $\Rightarrow \mu = 64$

Alternate Hypothesis $\Rightarrow \mu > 64$ (R.T.T)

$\alpha = 0.05$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{66 - 64}{3.162/\sqrt{10}} = 2$$



Z is in rejection region.

$\therefore$ Alternate Hypothesis is true $\Rightarrow$ The claim is true.

Q, A random sample from a company's very extensive files shows that the orders for a certain kind of machinery were filled respectively in 10, 12, 19, 14, 15, 18, 11 & 13 days. Use $\alpha = 0.01$ level of significance to test claim on avg such orders are filled in 10.5 days.

Ans $n = 8$ (small) $\bar{x} = \frac{10+12+19+14+15+18+11+13}{8} = 14$

$$S = \sqrt{\frac{16+4+25+0+1+16+9+1}{7}} = \sqrt{\frac{72}{7}} = 3.207$$

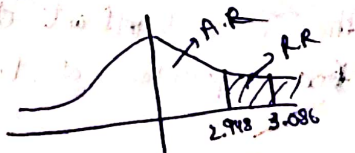
Null Hypothesis: $\mu = 10.5$

Alternate Hypothesis: $\mu > 10.5$ (R.T.T)

$\alpha = 0.01$

$$t = \frac{(\bar{x} - \mu) \sqrt{n}}{s} = \frac{(14 - 10.5) \sqrt{8}}{3.207} = 3.5 \sqrt{\frac{8}{729}} = 3.086$$

$$t_{0.01} = 2.998$$



Null Hypothesis is false.

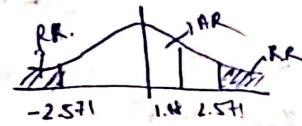
$\therefore$ Claim is false.

Q, A random sample of 6 steel beams has mean compressive strength of 58,392 PSI with std of 648 PSI. Use this information and the level of significance $\alpha = 0.05$ to test whether the true avg compressive strength of steel from which this sample came is 58000 PSI. Assume normality.

$n = 6$ (small) $\bar{x} = 58392$ $s = 648$ $t = 5$

N.H $\mu = 58000$
A.H $\mu \neq 58000$ (T.T.T).

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{392\sqrt{6}}{648} = 1.48179$$



Acceptance region

$\therefore$ claim is true.

Note: $t = \begin{cases} \frac{\bar{x} - \mu}{s/\sqrt{n}} & \text{when } s \text{ is calculated or not given} \\ \frac{\bar{x} - \mu}{\sigma/\sqrt{n-1}} & \text{when } s \text{ is given} \end{cases}$

Test of hypothesis for difference b/w 2 means large samples:-

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \text{ under Null Hypothesis.}$$

N.H $\mu_1 = \mu_2$

A.H $\mu_1 < \mu_2$ L.T.T
 $\mu_1 > \mu_2$ R.T.T
 $\mu_1 \neq \mu_2$ T.T.T

Q, In a survey of buying habits 400 women shoppers are chosen at random in supermarket A located in a certain section of the city. Their average weekly food expenditure is ₹ 250 with a std of ₹ 40. For 400 women shoppers chosen at random in supermarket B in another section of city. avg weekly food expenditure is ₹ 220 with std of ₹ 55. Test at 1% level of significance whether avg weekly food expenditure of two population of shoppers are equal.

$n_1 = 400$, $\bar{x}_1 = 250$, $s_1 = 40$, $n_2 = 400$, $\bar{x}_2 = 220$, $s_2 = 55$.

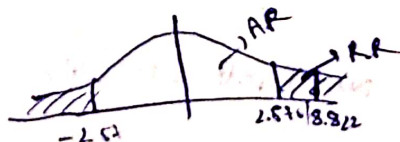
N.H $\Rightarrow \mu_1 = \mu_2$

A.H $\Rightarrow \mu_1 \neq \mu_2$ T.T.T

$\alpha = 0.005$

$Z_{\alpha/2} = 2.576$

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{30}{\sqrt{\frac{1600}{400} + \frac{3025}{400}}} = \frac{30 \times 20}{68.007} = 8.822$$



There is a significant difference b/w food expenditure of two population

N.H is false
claim is false.

Q. A simple sample of height of 6,400 English men has mean height of 67.85 inches and std of 2.56 inches while a simple sample of heights of 1600 Australians has mean of 68.55 inches and std of 2.52 inches. Do the data indicate Australian or an avg taller than English men?

Ans - English men.

$$n_1 = 6400$$

$$\bar{x}_1 = 67.85$$

$$s_1 = 2.56$$

Australian men

$$n_2 = 1600$$

$$\bar{x}_2 = 68.55$$

$$s_2 = 2.52$$

$$N.H = \mu_1 = \mu_2$$

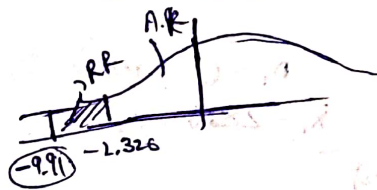
$$A.H = \mu_1 < \mu_2$$

L.T.T. (claim)

$$\text{Let } \alpha = 0.01$$

$$Z_{\alpha} = 2.326$$

$$Z = \frac{(\bar{x}_2 - \bar{x}_1) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{-0.7}{\sqrt{1.024 \times 10^{-3} + 3.96 \times 10^{-3}}} = \frac{-0.7 \times 100}{7.006 \times 10^{-2}} = \frac{-70}{7.006} = -9.991$$



Null Hypothesis is false

Alternate Hypothesis is true

∴ The data indicate Australians on an avg are taller than English men.

Q. A company claims that alloying reduces resistance of an electric wire by more than 0.05 ohms. To test this claim samples of alloyed wire and samples of standard wire are tested which have given following results

Types of wires	size	mean	std
Standard	32	0.136	0.004
alloyed	32	0.083	0.005

std

$$n_1 = 32$$

$$\bar{x}_1 = 0.136$$

$$s_1 = 0.004$$

alloyed

$$n_2 = 32$$

$$\bar{x}_2 = 0.083$$

$$s_2 = 0.005$$

$$N.H = \mu_1 - \mu_2 = 0.05$$

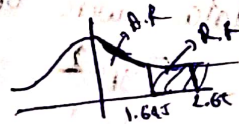
$$A.H = \mu_1 > \mu_2 \text{ claim (R.T.T.)}$$

$$\alpha = 0.05$$

$$Z_{0.05} = 1.645$$

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(0.136 - 0.083) \times 10^3}{\sqrt{\frac{16}{32} + \frac{25}{32}}} = \frac{53}{1.1319} = 46.83$$

$$= 2.65$$



NH is false

Alternate hypothesis is true.

alloying reduces resistance of an electric wire by more than 0.05 ohms

Test of Hypothesis for diff 2 means small samples

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ under N.H with } \nu = n_1 + n_2 - 2 \text{ D.O.F}$$

where $s = \begin{cases} \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} & \text{when } s_1 \text{ and } s_2 \text{ are given} \\ \sqrt{\frac{\sum (x_{1i} - \bar{x})^2 + \sum (x_{2i} - \bar{x}_2)^2}{n_1 + n_2 - 2}} & \text{when } s_1 \text{ and } s_2 \text{ are calculated not given} \end{cases}$
(Pooled standard deviations)

Q. 3 Two horses A & B were tested according to time in sec to run a particular track with following results. Test whether two horses have same running capacity

A	28	30	32	33	33	29	34
B	29	30	30	24	27	29	

Ans: $n_1 = 7$ (small) $n_2 = 6$ (small)

$\bar{x}_1 = 31.2857 = \mu_1$ $\bar{x}_2 = 28.16 = \mu_2$

$$s = \sqrt{\frac{\sum (x_1 - \bar{x})^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}}$$

$\sum (x_1 - \bar{x})^2 = 0.7569 + 3.3856 + 3.3856 + 17.3056 + 1.3456 + 0.7056 = 26.88$

$$= \sqrt{\frac{31.42 + 26.88}{11}} = \sqrt{5.30} = 2.304$$

N.H $\Rightarrow \mu_1 = \mu_2$

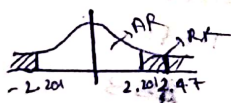
A.H $\Rightarrow \mu_1 \neq \mu_2$ (T.T.T)

L.O.S $\Rightarrow \alpha = 0.05$

$\alpha/2 = 0.025$

$t_{\alpha/2, \nu} = 2.201$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{3.1257}{2.304 \sqrt{\frac{1}{7} + \frac{1}{6}}} = \frac{3.1257}{2.304 \times 0.55} = \frac{3.1257}{1.265} = 2.47$$



Reject Null Hypothesis

They don't have same running capacity.

Q. Samples of two types of electric bulbs were tested for length of life and following data were obtained.

Type 1

$n_1 = 8$

$\bar{x}_1 = 1234 \text{ hrs}$

$s_1 = 36 \text{ hrs}$

Type 2

$n_2 = 7$

$\bar{x}_2 = 1036 \text{ hrs}$

$s_2 = 40 \text{ hrs}$

Is the difference in means sufficient to warrant that type 1 is superior to type 2 regarding length of life

Ans: $n_1 = 8$ $n_2 = 7$

$\bar{x}_1 = 1234 \text{ hrs}$ $\bar{x}_2 = 1036 \text{ hrs}$

$s_1 = 36 \text{ hrs}$ $s_2 = 40 \text{ hrs}$

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{10,368 + 11,200}{13}} = \sqrt{1659.0769} = 40.73$$

$$N.H \Rightarrow \mu_1 = \mu_2$$

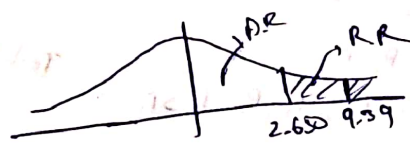
$$A.H \Rightarrow \mu_1 > \mu_2 \text{ (R.T.T.)}$$

$$\text{let } \alpha = 0.01$$

$$t_{\alpha, n} = t_{0.01, 13} = 2.650$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{198}{40.73 \sqrt{\frac{1}{8} + \frac{1}{7}}} = \frac{198}{40.73 \times 0.539} = 9.39$$



Null Hypothesis is false

claim is true.

Test of Hypothesis for single proportion ~~small~~ large sample

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \Rightarrow Z = \frac{\bar{p} - p}{\sqrt{pq/n}}$$

Null Hypothesis $H_0 \Rightarrow P = p_0$

Alternate Hypothesis $H_1 \Rightarrow P \neq p_0$ (T.T.T.) $P > p_0$ (R.T.) $P < p_0$ (L.T.)

Test statistic $Z = \frac{\bar{p} - p}{\sqrt{pq/n}}$

Q, In a big city 325 men out of 600 men were found to be smokers. Does this information support conclusion that majority of men in city are smokers

Ans:- Let X represents

smokers. $\Rightarrow x = 325$ $n = 600$

$$\bar{p} = \frac{x}{n} = \frac{325}{600} = 0.54$$

$$N.H: H_0: P = 0.5$$

$$A.H: H_1: P > 0.5$$

$$\text{let } \alpha = 0.05$$

$$Z_{0.05} = 1.645$$

$$q = 0.5$$

$$Z = \frac{\bar{p} - p}{\sqrt{pq/n}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = \frac{0.04 \times \sqrt{600}}{0.5} = \frac{0.979}{0.5} = 1.959$$



Null Hypothesis is false
claim is True

Q, A manufacturer claimed that atleast 95% of the equipment which he supplied to a factory confirmed to specification. An examination of a sample of 200 pieces of the equipment revealed that 18 were faulty. Test his claim at 5% level of significance.

Let X represent faulty $\Rightarrow x = 18$ $n = 200$

$$\bar{p} = \frac{x}{n} = \frac{18}{200} = 0.09$$

$$\alpha = 0.05$$

$$Z_{0.05} = 1.645$$

$$N.H: H_0: P = 0.95$$

$$A.H: H_1: P < 0.95 \text{ (L.T.T.)}$$

$$Z = \frac{\bar{p} - p}{\sqrt{pq/n}} = \frac{0.09 - 0.95 \times \sqrt{200}}{\sqrt{0.95 \times 0.05}} = \frac{0.5656}{\sqrt{0.2199}} = 2.5956$$



N.H is false
claim is rejected. true rejected

Test of Hypothesis for two proportions

Same formula, just replace z with t , z_{α} with t_{α}

Model 1

NH: $H_0: P_1 = P_2$

Alt: $H_1: P_1 \neq P_2, P_1 < P_2, P_1 > P_2$

Test statistic:

$$Z = \frac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Where $\hat{P} = \frac{x_1 + x_2}{n_1 + n_2}$ (or) $\frac{n_1 \bar{P}_1 + n_2 \bar{P}_2}{n_1 + n_2}$

Q. Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of proposal. Test the hypothesis that the proportions of men and women in favour of proposals are same. Test at 5% level.

men
 $n_1 = 400$
 $x_1 = 200$
 $\bar{P}_1 = \frac{x_1}{n_1}$
 $\bar{P}_1 = 0.5$

women
 $n_2 = 600$
 $x_2 = 325$
 $\bar{P}_2 = \frac{x_2}{n_2}$
 $\bar{P}_2 = 0.5416$

population proportion P_1 = population proportion P_2
NH $H_0: P_1 = P_2$ (same)
Alt $H_1: P_1 \neq P_2$ (T.T.) (not-same)

$\alpha = 0.05$
 $\alpha/2 = 0.025$

$z_{\alpha/2} = 1.96$

$\hat{P} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{200 + 325}{400 + 600}$
 $\hat{P} = 0.525$

$Z = \frac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$
 $Z = \frac{(0.5 - 0.5416) - 0}{\sqrt{0.525(0.475)\left(\frac{1}{400} + \frac{1}{600}\right)}}$

$= \frac{-0.0416 \sqrt{1400}}{\sqrt{(0.525)(0.475)}} = \frac{-0.0416 \times 37.416}{0.499} = -1.29$



Acceptance region

Null Hypothesis is true

Claim is acceptable

Q. In a city A, 20% of a random sample of 900 school boys have a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys has the same defect. Is the difference b/w proportions significant at 0.05 level of significance.

$n_1 = 900$
 $\bar{P}_1 = 0.20$

$n_2 = 1600$
 $\bar{P}_2 = 0.185$

N.H $H_0: P_1 = P_2$

Alt $H_1: P_1 \neq P_2$ (T.T.)

$\alpha = 0.05 \Rightarrow \alpha/2 = 0.025$

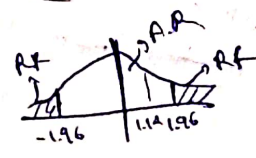
$z_{\alpha/2} = 1.96$

$\hat{P} = \frac{n_1 \bar{P}_1 + n_2 \bar{P}_2}{n_1 + n_2} = 0.19$

$\hat{Q} = 0.81$

$Z = \frac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

$= \frac{0.015}{\sqrt{(0.19)(0.81)\left(\frac{1}{900} + \frac{1}{1600}\right)}} = \frac{0.015 \sqrt{14400}}{\sqrt{(0.19)(0.81)(25)}} = \frac{(20)(0.015)}{5(0.323)} = 1.14$



NH is true.
No difference

Model - II

$$N.H H_0: P_1 - P_2 = 8$$

$$A.H H_1: P_1 - P_2 \neq 8$$

$$Z = \frac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

Q. A cigarette manufacturing firm claims that its brand A line of cigarettes outsells its brand B by 8%. It is found that 42 out of sample of 200 smokers prefer brand A and 18% of another sample of 100 smokers prefer brand B. Test whether 8% difference is a valid claim

$$n_1 = 200$$

$$x_1 = 42$$

$$P_1 = \frac{42}{200} = 0.205$$

$$n_2 = 100$$

$$x_2 = 18$$

$$P_2 = 0.18$$

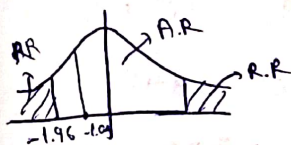
$$N.H H_0: P_1 - P_2 = 0.08$$

$$A.H H_1: P_1 - P_2 \neq 0.08$$

$$\alpha = 0.05$$

$$Z_{\alpha} = 1.96$$

$$Z = \frac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}} = \frac{0.03 - 0.08}{\sqrt{\frac{0.205 \times 0.79}{200} + \frac{0.18 \times 0.82}{100}}} = -1.04$$



Accept null Hypothesis
claim is true.

Model - III

$$Z = \frac{P_1 - P_2 - (\bar{P}_1 - \bar{P}_2)}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

$$N.H H_0: \bar{P}_1 = \bar{P}_2$$

$$A.H H_1: \bar{P}_1 \neq \bar{P}_2$$

Q.

$$\text{Ans: } n_1 = 1200, n_2 = 900, \alpha = 0.05$$

$$P_1 = \frac{x_1}{n_1} = 0.3, P_2 = \frac{x_2}{n_2} = 0.25$$

$$\text{Null Hypothesis } H_0: \bar{P}_1 = \bar{P}_2$$

$$\text{Alternate Hypothesis } H_1: \bar{P}_1 \neq \bar{P}_2 \text{ (I.T.T.)}$$

$$\alpha = 0.05$$

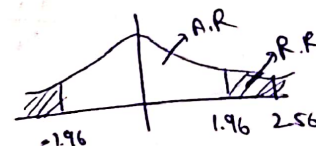
$$Z_{\alpha} = 1.96$$

$$Z = \frac{(P_1 - P_2) - (\bar{P}_1 - \bar{P}_2)}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}} = \frac{0.3 - 0.25}{\sqrt{\frac{0.3(0.7)}{1200} + \frac{0.25(0.75)}{900}}} = 2.56$$

$$Z = 2.56$$

N.H is false

difference is significant.



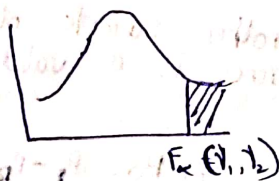
F-Test:-

Here no two tailed test

To test whether 2 population variances are equal or not

N.H $H_0: \sigma_1^2 = \sigma_2^2$

A.H $H_1: \sigma_1^2 \neq \sigma_2^2$



$$F = \begin{cases} \frac{S_1^2}{S_2^2} \text{ provided } S_1^2 > S_2^2 \text{ when } S_1^2 \text{ \& } S_2^2 \text{ are not given} \\ \frac{\frac{n_1}{n_1-1} S_1^2}{\frac{n_2}{n_2-1} S_2^2} \text{ where } S_1^2 \text{ \& } S_2^2 \text{ are given} \end{cases}$$

$v_1 = n_1 - 1 \quad v_2 = n_2 - 1$

Q, In one sample of 8 observations. the sum of squares of deviations of sample values from the sample mean was 84.4 and in the other sample of 10 observations it is 102.6. Test whether difference is significant at 5%.

Ans: $n_1 = 8 \quad n_2 = 10$

$$S_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1}$$

$$S_2^2 = \frac{\sum (x_i - \bar{x})^2}{n_2 - 1}$$

$$S_1^2 = \frac{84.4}{7} = 12.057$$

$$S_2^2 = \frac{102.6}{9} = 11.4$$

$$S_1^2 > S_2^2$$

N.H $H_0: \sigma_1^2 = \sigma_2^2$

A.H $H_1: \sigma_1^2 \neq \sigma_2^2$

$\alpha = 0.05$ (~~0.01~~)

$F_{\alpha}(7, 9) = 3.29$

$$F = \frac{S_1^2}{S_2^2} = 1.057$$

$F < F_{\alpha}(7, 9)$

A.R. N.H. true

Q, Pumpkins were grown under two experimental conditions. 2 random samples of 11 & 9 pumpkins shows that sample std of their weights $S_1 = 0.4$ and $S_2 = 0.5$ respectively. Assuming weight dist are normal. Test hypothesis that variances are equal.

Ans: $n_1 = 11 \quad n_2 = 9$

$S_1 = 0.4 \Rightarrow S_1^2 = 0.16$

$S_2 = 0.5 \Rightarrow S_2^2 = 0.25$

Let $\alpha = 5\%$

A.H $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

$F_{\alpha}(10, 8) = 3.35$

$$F = \frac{\frac{n_1}{n_1-1} S_1^2}{\frac{n_2}{n_2-1} S_2^2} = \frac{\frac{11}{10} \times 0.16}{\frac{9}{8} \times 0.25}$$

$F < F_{\alpha}$

$$= \frac{11 \times 0.16 \times 8}{10 \times 9 \times 0.25} = \frac{14.08}{22.5} = 0.625$$

Q, The nicotine contains in mg. in 2 samples of tobacco were found to be as follows. ~~Sample A~~

Sample A	29	27	26	24	25	-
Sample B	27	30	28	31	24	36

population

can it be said that the two samples came from same normal

Ans: $n_1 = 5$ $n_2 = 6$.

$$\bar{x}_1 = \frac{24+27+26+24+25}{5}$$

$$= 24.6$$

$$\bar{x}_2 = 29.$$

$$S_1^2 = \frac{21.2}{4} = 5.3$$

$$S_2^2 = \frac{4+1+1+4+4+4}{5} = 2.8$$

$$S_2^2 = 21.6$$

$$\mu_1$$

$$\mu_2$$

$$N.H. H_0: \sigma_1^2 = \sigma_2^2$$

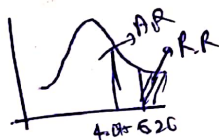
$$A.H. H_1: \sigma_1^2 \neq \sigma_2^2$$

$$L.O.S \alpha = 0.05$$

$$S_2^2 > S_1^2$$

$$f_{0.05}(5,4) = 6.26.$$

$$f = \frac{S_2^2}{S_1^2} = \frac{21.6}{5.3} = 4.075.$$



Accept Null Hypothesis

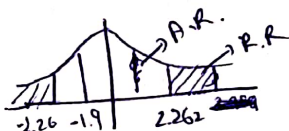
$$N.H. H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2 \text{ (T.T.T)}$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$t_{\alpha/2} = 19.02 \rightarrow 2.262$$



Accept Null Hypothesis

both samples are from same population.

χ^2 test as a goodness of fit

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}, \text{ where 'k' is no. of cells or classes.}$$

' O_i ' is observed sample frequency
' E_i ' is expected frequency.

χ^2 is a random variable with 'v' degree of freedom.